# MULTIPLICITY ONE FORCES TRANSVERSAL INTERSECTION: E = 1 AND SMOOTHNESS IN GENERAL POINTS

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

*Source reference.* Original section 0AZA: sources/stacks-chow/tags.txt, source lines 13818–13818. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 00KU: sources/stacks-chow/tags.txt, source lines 764–764. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 00NQ: sources/stacks-chow/tags.txt, source lines 868–868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 00NP: sources/stacks-chow/tags.txt, source lines 867–867. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 00KW: sources/stacks-chow/tags.txt, source lines 766–766. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 062F: sources/stacks-chow/tags.txt, source lines 7794–7794. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $X$ be a nonsingular variety. Let $V, W \subset X$ be closed subvarieties which intersect properly. Let $Z$ be an irreducible component of $V \cap W$ and assume that the multiplicity (in the sense of Section 0AZA) of $Z$ in the closed subscheme $V \cap W$ is 1. Then $e(X, V \cdot W, Z) = 1$ and $V$ and $W$ are smooth in a general point of $Z$.*

*Date*: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* Let $(A, \mathfrak{m}, \kappa) = (\mathcal{O}_{X,\xi}, \mathfrak{m}_\xi, \kappa(\xi))$ where $\xi \in Z$ is the generic point. Then $\dim(A) = \dim(X) - \dim(Z)$, see Varieties, Lemma 0A21. Let $I, J \subset A$ cut out the trace of $V$ and $W$ in $\mathrm{Spec}(A)$. Set $\overline{I} = I + \mathfrak{m}^2/\mathfrak{m}^2$. Then $\dim_\kappa \overline{I} \leq \dim(X) - \dim(V)$ with equality if and only if $A/I$ is regular (this follows from the lemma cited above and the definition of regular rings, see Algebra, Definition 00KU and the discussion preceding it). Similarly for $\overline{J}$. If the multiplicity is 1, then $\mathrm{length}_A(A/I + J) = 1$, hence $I + J = \mathfrak{m}$, hence $\overline{I} + \overline{J} = \mathfrak{m}/\mathfrak{m}^2$. Then we get equality everywhere (because the intersection is proper). Hence we find $f_1, \ldots, f_a \in I$ and $g_1, \ldots g_b \in J$ such that $\overline{f}_1, \ldots, \overline{g}_b$ is a basis for $\mathfrak{m}/\mathfrak{m}^2$. Then $f_1, \ldots, g_b$ is a regular system of parameters and a regular sequence (Algebra, Lemma 00NQ). The same lemma shows $A/(f_1, \ldots, f_a)$ is a regular local ring of dimension $\dim(X) - \dim(V)$, hence $A/(f_1, \ldots, f_a) \to A/I$ is an isomorphism (if the kernel is nonzero, then the dimension of $A/I$ is strictly less, see Algebra, Lemmas 00NP and 00KW). We conclude $I = (f_1, \ldots, f_a)$ and $J = (g_1, \ldots, g_b)$ by symmetry. Thus the Koszul complex $K_\bullet(A, f_1, \ldots, f_a)$ on $f_1, \ldots, f_a$ is a resolution of $A/I$, see More on Algebra, Lemma 062F. Hence

$$\begin{aligned} \mathrm{Tor}_p^A(A/I, A/J) &= H_p(K_\bullet(A, f_1, \ldots, f_a) \otimes_A A/J) \\ &= H_p(K_\bullet(A/J, f_1 \bmod J, \ldots, f_a \bmod J)) \end{aligned}$$

Since we've seen above that $f_1 \bmod J, \ldots, f_a \bmod J$ is a regular system of parameters in the regular local ring $A/J$ we conclude that there is only one cohomology group, namely $H_0 = A/(I + J) = \kappa$. This finishes the proof. $\square$